

Fall 2025 DRP Presentations

Thursday, November 20th, East Hall

Introduction

Welcome to the Fall 2025 DRP Presentations. We will have one room this time, so you get to see everyone! The abstracts for all talks are on the following pages, followed by abstracts for the final papers. Please feel free to invite other math students, family, and friends.

Session 1A: 5:00 – 6:00pm East Hall Room 2866

Sophia Guo (She/Her)	<i>All Aboard the Symmetry Plane</i>
Taran Knuston (She/Her)	<i>Modules and Hom/Tensor Functors</i>
Albert Ye (He/Him)	<i>Definition of Distribution Functions</i>
James Kluge (He/Him)	<i>E-Fish-ent Scaling of RL Algorithms</i>
Dennis Farmer (He/Him)	<i>Numerical Methods for Fluid Flow</i>

Pizza Break+ Paper Presentation: 6:00 – 7pm East Hall Upper Atrium

Xiangnong Wu	<i>Introduction to Field Theory</i>
Austin Moon (He/Him)	<i>Intersecting Parallel Lines at Infinity: Explorations in Projective Geometry</i>
Yawen Liu	<i>Random Lozenge Tilings</i>
Laith Amjad (He/Him)	<i>Spectral Graph Theory</i>
Chloe Eberwein (She/Her)	<i>Orthogonal Eigenfunctions and Series Expansions in PDE Solutions</i>
Jay Trombley	<i>On Bezout's Theorem for Plane Curves</i>

Abstracts (Presentations)

Sophia Guo: *All Aboard the Symmetry Plane*

Mentor: Veronica Lang

Group theory arises from the study of symmetry and structure across mathematics and the natural sciences. In this presentation, we will begin with some basic definitions in group theory and then characterize finite plane symmetry groups. We will conclude with a brief discussion of some special infinite plane symmetry groups, including discrete frieze groups and crystallographic groups.

Taran Knutson: *Modules and Hom/Tensor Functors*

Mentor: Gahl Shemy

This talk will discuss modules as a natural idea that encompasses vector spaces, abelian groups, and ideals. It will contain examples and basic statements about the covariant and contravariant Hom functors on $R\text{-Mod}$ as well as the tensor product. Then we will move to an intuitive description of Tensor-Hom adjunction.

Chenzhang (Albert) Ye: *Definition of Distribution Functions*

Mentor: Horacio Moreno Montanes

Based on Essentials of Stochastic Processes by Richard Durrett, my presentation will introduce the random variable, its difference under Z and R, and an exponential distribution with rate λ . Then explaining density function, as derivative of the distribution function with graphs. Using Expected Value ET, we explain how continuous random variables differentiate from discrete rv. If time allows, we'll introduce Lack of Memory Property.

James Kluge: *E-fish-ent Scaling of RL Algorithms*

Mentor: Aditya Priya

Reinforcement learning algorithms are often presented as autonomous systems that learn from experience, but their success hinges critically on human design choices. This presentation demonstrates how an engineer's conceptualization of the problem – in our case, a humble penguin catching fish – can drastically alter the complexity, training time, and accuracy of our models. I show that moving from raw positional states to abstracted relative features improves success rates from 35% to 100% with only 1/10th of the prior training time. Moreover, the abstract algorithm scales much better with increased problem complexity, showing that effective AI design is as dependent on human conceptualization as it is on computational power.

Dennis Farmer: *Numerical methods for fluid flow*

Mentor: Kausik Das

Estimating physical parameters from experimental data is a problem of great importance in the study of fluid flows. After introducing the Navier-Stokes equations for incompressible flow, we derive Couette flow as a concrete example of a steady-state solution around which linearization is possible. We investigate a computational approach for recovering the underlying system matrix from simulated streamline data from a linear flow field, examining the loss function landscape when the matrix is defective or ill-conditioned, as well as the effects of short-time data.

Abstracts (Papers)

Xiangnong Wu: *Introduction to Field Theory*

Mentor: Simran Khunger

In this expository paper, we explore some basic tools and facts for studying field extensions, focusing on examples. In particular, we will cover construction of field extensions, degree of extensions, extensions generated by one or more elements, algebraic extensions, splitting fields, algebraic closures, and separable and inseparable extensions. Those topics are important for studying Galois's theory, which is essential in algebraic number theory.

Austin Moon (He/Him): *Intersecting Parallel Lines at Infinity: Explorations in Projective Geometry*

Mentor: Dawei Shen

Projective geometry is the study of geometric properties invariant under projective transformations. Geometric objects are embedded in the projective spaces, which can be understood as compactifications of affine spaces by adding objects at infinity. In order to explore projective geometry over $\mathbb{R}$, we define projection maps to motivate the definition of projective spaces in terms of homogeneous projective coordinates.

Under this framework, we discuss results such as Desargue's Theorem and Pappus's Theorem. In particular, we illustrate how projective spaces give rise to a suitable ambient space in which geometric statements can be proved algebraically.

Yawen Liu: *Random Lozenge Tilings*

Mentor: Zach Deiman

The paper will be a brief introduction of Random Lozenge tiling models, based on the lecture notes by Professor Vadim Gorin. First we will introduce lozenge tiling on a triangular domain, and Kastelyn matrix that can help to compute the total number of tilings. With this, we may construct a probability measure on tilings. Finally, the paper will end by defining height function and stepped surfaces, and a result of the limit shape.

Laith Amjad (He/Him): *Spectral Graph Theory*

Mentor: Yixiang Fei

In this exploration, we look at the connection between Linear Algebra and Graph Theory via the Adjacency and Laplacian matrix. We first introduce some basic facts about graphs and results which help us establish facts about the eigenvalues of irreducible matrices. We continue by exploring properties of the friendship graphs using the eigenvalues of the adjacency matrix. Finally, we look at the number of spanning trees of a graph purely as a function of its spectrum.

Chloe Eberwein (She/Her): *Orthogonal Eigenfunctions and Series Expansions in PDE Solutions*

Mentor: Maria Madrugo

The use of orthogonal eigenfunctions forms the mathematical backbone of many solution techniques for partial differential equations. Beginning with the derivation of eigenvalue problems from separation of variables, this paper explores how certain differential operators have real eigenvalues, along with a basis of mutually orthogonal eigenfunctions. These properties enable the representation of complex physical states, such as temperature distributions and wave vibrations, as infinite series expansions in sine, cosine, or other orthogonal bases. Thus, the completeness of these eigenfunction sets ensures that any function satisfying the boundary conditions can be represented as a convergent series expansion, providing a powerful framework for solving PDE's.

Jay Trombley: *On Bezout's Theorem for Plane Curves*

Mentor: Ben Baily

This paper discusses how to count points of intersection between algebraic curves using Bezout's Theorem. We cover concepts central to Bezout's Theorem such as the projective plane, parametrization of curves, and multiplicities of intersection points over an algebraically closed field. Relevant elementary examples will demonstrate the concepts, and will assist in visualization of the connection between the degrees of plane curves and their intersections.